Jaejin Cha

Software Engineer | ML Systems, Simulation & Planning
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Summary

Software engineer and M.S. Computer Science student with AI/data systems experience and research in autonomous systems. IROS 2026 author in multi-robot planning, with autonomous-driving simulation work in reinforcement and imitation learning.

Technical Skills

Languages: Python, C++, SQL; **Systems & Data:** Linux, Docker, Git, REST APIs, PostgreSQL, Apache Spark, HDFS, Oracle Database, OCI
ML & Planning: PyTorch, Gymnasium, reinforcement learning, imitation learning, Graphs of Convex Sets, multi-agent planning

Experience

Hewlett Packard Enterprise — **Spring, TX**
Software Engineer – AI Applications and Enterprise Systems — Intern: May 2025 – May 2026 | Full-time: Jun 2026 – Present

- Build production AI applications enabling operations teams to explore enterprise data through natural-language queries and generate on-demand insights.
- Implement text-to-SQL and tool-calling workflows using Oracle AI Database and PL/SQL packages, incorporating business-rule constraints.
- Built a custom vector storage and similarity-search layer in Oracle 19c using PL/SQL, REST APIs, and embedding models.

Texas A&M University — **College Station, TX**
Graduate Researcher – Multi-Robot Planning & Coordination — Aug 2024 – Present

- Formulated cyclic multi-robot rendezvous as constrained shortest-path planning on a Graph of Convex Sets in phase–time space, enforcing timing and traversal-frequency feasibility across range-limited communication events.
- Built Python simulation and GCS planning infrastructure for multi-robot coordination; implemented graph sparsification to reduce planning complexity, evaluated against consensus baselines on an 11-robot topology, and validated trajectories on Robotarium hardware.

Texas A&M University – NIH AIM-AHEAD Research — **Remote**
Research Assistant — Aug 2025 – Present

- Built Python/SQL pipelines over the OCHIN 2023 EHR dataset to construct a 500,000+ T2D patients cohort for cardiovascular risk analysis.
- Evaluated AHA PREVENT risk equations using C-index, calibration, and subgroup fairness analyses across the clinical cohort.

ENDEAVR Institute — **College Station, TX**
Machine Learning Engineer Intern — Oct 2024 – Feb 2025

- Implemented a PPO-based generator for a GAIL imitation-learning pipeline in PyTorch and validated it on Gymnasium environments.

University of Delaware — **Newark, DE**
Autonomous Driving Research Intern — Jun 2023 – Aug 2023

- Modified a Gymnasium-based highway simulator and implemented DQN and DDPG agents for discrete and continuous vehicle control.
- Ran reward-weight experiments on trade-offs between energy efficiency and driving metrics; presented results at a UD research symposium.

Texas A&M University — **College Station, TX**
Undergraduate Research Assistant — Sep 2022 – May 2023

- Configured Apache Spark and HDFS environments on Linux clusters and developed Python/Bash tooling to collect and analyze garbage-collection logs for runtime and memory profiling.
- Modified Peregrine source code to convert multi-label graph datasets into binary format for graph-processing workflows.

Publications

Jaejin Cha, Dylan Shell. *"Planning and Execution for Multi-Robot Cyclic Rendezvous via Graphs of Convex Sets"*. IROS 2026 (accepted).
Jaejin Cha. *"Rendezvous-Based Multi-Robot Communication Under Cyclic Paths"*. Master's thesis (defended), 2026.

Selected Projects & Activities

AutonomyForge | C++20, Python — Aug 2026 – Present

- Designing and building a C++20 multi-agent simulation and planning framework for deterministic autonomy experiments, with modular interfaces for scenarios, planners, simulation, and evaluation.

Bitcoin Trading Simulator | Python, PyTorch, NumPy — Nov 2024 – Feb 2025

- Built an MDP-based Bitcoin trading environment from Kaggle data and trained reinforcement-learning agents for sequential decisions.

Education

Texas A&M University — **College Station, TX**
M.S. Computer Science, GPA: 3.71/4.00 - Thesis in Multi-Robot Systems | Advisor: Dr. Dylan Shell — Expected Dec 2026

Texas A&M University — **College Station, TX**
B.S. Computer Science, Minor in Mathematics — Aug 2024